myjournal manuscript No.
(will be inserted by the editor)

Semantic Model for Adaptive and Argument Based Human-Agent Dialogues

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Received: date / Accepted: date

Abstract —— A common conversation between an older adult and a nurse about health-related issues includes topics such as troubles with sleep, reasons for walking around nighttime, pain conditions, etc. Such a dialogue can be regarded as a "natural" dialogue emerging from the participating agents' lines of thinking, their roles, needs and motives, while switching between topics as the dialogue unfolds. The purpose of this work is to define a generic model of purposeful human-agent dialogue activity including different types of argumentation dialogues, suitable for health-related topics. This is done based on analyses of scenarios, personas and models of human behavior. The results include four models, which need to be included in a software agent's belief base; i) a user model, ii) a model of the domain knowledge related to the topic of the dialogue, iii) a dialogue activity model, and iv) an agent model. This article specifies the dialogue activity model, which were implemented into a prototype system for human-agent dialogues. The dialogue system was evaluated by therapists and a group of older adults.

Keywords Personalization · Dialogue systems · Health promotion · Argumentation · Intelligent agent · User model

1 Introduction

Assistive technology aims to support an individual in accomplishing activities, which they need to be able to do in the presence of decreased functionality or ability. In this work the definition of *active* assistive technology used by Kennedy and co-workers [1] is applied to distinguish systems, which includes automated processing of health information during a human agent's interaction with the system and which may output tailored responses to the human agent in the process. The

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purposes of active assistive technology include the following: to increase knowledge, assist in deciding about actions to make, and promote changes of unhealthy behavior (i.e., in the form of *behavior change systems*).

Our work focuses on dialogues between a human actor and an active assistive technology in the form of an intelligent software agent. The concept of Embodied Cognitive Agents (ECA) is generally used for such systems. The ECA uses a virtual representation of a human with the ability to send information through body language in addition to linguistic messages. However, we restrict the focus in this work to structured linguistic dialogues [**?**,**?**] based on semantic models of relevant knowledge.

The aim is to enable dialogues between a human agent and a software agent, which are tailored to the human's needs, preferences and goals. For this purpose, the software agent needs the following capabilities, which has to different extent also been described in literature (e.g., [2–6]):

1. Being autonomous (e.g., decides upon actions to make, takes initiatives in order to reach a goal),
2. Utilize a *semantic model* shared between the human and the agent for communication and knowledge exchange purposes,
3. Generate new knowledge and learn,
4. Being cooperative,
5. Being able to deal with affective components and topics in a dialogue,
6. Being adaptive to the human's needs, preferences and the situation.

The goal is to develop and combine the above-mentioned capabilities to build a software agent that interacts with the human as their personal coach, friend or a discussion partner, i.e., as a *Coach Agent* as described in [7]. In this work, we focus on the dialogue activity, and restrict the agent's autonomy to the agent's task of selecting topics, dialogue types and moves within a dialogue initiated by the human.

Typically, a team of healthcare professionals can be involved in an elderly person's care, such as doctors, nurses and occupational therapists. These together with the human cooperate to gain optimal understanding of the human's health condition in a process of *assessment*. Consequently, there is a need for combining information from different sources to understand the external factors affecting the human. The support provided to the human is aimed to be person-centered, and the assessment includes deciding upon what *interventions* may be implemented for improving the human's situation.

In this paper, we focus the assessment, while leaving actuating intervention to future work. We focus on the knowledge, which the software agent needs to have and obtain for contributing to the dialogue, and for behaving in a purposeful way. In particular, we are interested in how a dialogue, which includes the types of argumentation dialogues information-seeking, inquiry (generate new knowledge), deliberation (deciding about what actions to do) and persuasion dialogues defined by Walton and Krabbe [8] can be organized following theories of purposeful human activity (Activity Theory [9] and Self-Determination Theory [**?**]). The main contribution of this paper is a conceptual and formal model of the Coach Agent's knowledge regarding conducting the dialogue activity for assessment purposes.

A prototype dialogue system was implemented, which enables the human actor to conduct dialogues on different health-related topics with the Coach Agent in their home environment. The prototype was evaluated in a pilot study involving a group of therapists. The purpose was to evaluate from healthcare experts' perspectives, the agent's behavior when conducting the dialogue, its use of the domain knowledge and user model.

This paper is organized as follows. In the following section the methods are described and in Section 3 an analysis of related work is provided. Section 4 introduces the use case scenario of an older woman and the results of the analysis of the scenario, which provided requirements and motivations for our design. In Section 5 a model of the dialogue activity is provided. Section 6.6 describes the pilot evaluation study. The results are discussed and the article ends with conclusions and comments on future work.

2 Methods

A literature study was conducted where the following topics were studied: i) the agent's role, ii) the purpose of dialogue systems, iii) the knowledge model of the agent, the user, the domain and the dialogue activity, iv) who develops the knowledge model, v) sources for the knowledge, vi) representation format and how generic the representation is, and vii) issues regarding the dialogue execution.

A semantic model for adaptive human-agent dialogues is designed based on the persona and scenario of a female older adult named Eva, and the dialogues aimed for supporting a human actor described in [7,10]. The scenario was analysed, providing baseline requirements and a conceptual model of goal-directed dialogue activity.

The theoretical base for analysis is Activity Theory [9], which also informed the models generated as results of our work.

A pilot evaluation study was conducted involving a group of female professionals in occupational therapy and physiotherapy, specialized in the needs of older adults. The study was formative, with the results aimed to inform further development.

The main research question addressed in this paper is that, can the use of personalised argumentation dialogues with older adults on health-related topics supplement to the healthcare professionals future intervention?

To answer this research question, we design and develop a goal-directed semantic dialogue activity model. This model was evaluated by professionals in occupational therapy and physiotherapy. The sub-research questions, to answer the main research question, were the following:

1. Was the outcome of each argumentation dialogue satisfying the requirements expected by the professionals?
2. What new information can be added to the knowledge base?
3. Which features of the user interaction can be added or enhanced?

During the evaluation study, the healthcare professional tested the dialogue system by initiating a dialogue with the Coach agent. As the dialogue unfolded, the professional's dialogue moves on the screen and their running comments were observed and noted down. When the evaluator observed the end of each type of dialogue, asked the following questions:

1. End of Information-seeking dialogue
   (a) Were the questions asked by the agent considered as appropriate to the topic?
   (b) Was there any other question the professional would like the agent to ask?
   (c) Was the user-interaction design for answering the questions intuitive?
2. End of Inquiry dialogue
   (a) Did the professional and agent find a new knowledge about the older adult's health condition?
3. End of Deliberation dialogue
   (a) Was there a plan of action agreed upon between the participants of the dialogue?
   (b) Any new actions that can be added to the knowledge base?

Observation of use and open-ended interviews were the methods used for collecting data. The data was analysed qualitatively using content analysis.

3 Related Work

Research literature shows an increasing number of applications of software agents that interact with the human actors for health-related purposes (e.g., [11, 12, 5, 13–15]). A review of behavior change systems utilizing personalization technologies for accomplishing active assistance is provided in [1]. It was concluded that an agent provides an effective interface modality especially for the applications that require repeated interactions over longer period of time, which is crucial for applications supporting behavior change [1, 16].

One example is the system, which has been developed for older adults with cognitive impairment described in [11]. However, it focuses mainly on generating reminders about the activities of their daily living and takes no part in a complex dialogue with the user. Agent-based systems which interact with human actors through dialogues are less common. They are typically developed for a specific task or for a limited domain, and the dialogues are tested in specialized environments [15, 13].

Bickmore et. al [15] developed an animated conversational agent that provides information about a patient's hospital discharge plan in an empathic fashion. The patients rated the agent very high on measures of satisfaction and ease of use, and also as being the most preferred over their doctors or nurses in the hospital when receiving their discharge information. This was related to the agent's capability to adapt to the user's pace of learning and giving information in a nonjudgmental manner. Another reason for the positive results was the amount of time that the agent spent with users helped them to establish a stronger emotional bond.

In the counseling dialogue system discussed in [16], the agent performs motivational interviewing with its knowledge about behavioral medicine and expresses empathy in dialogues. It provides the human agent with advice about exercise and healthy actions and follows up by asking the human followup questions. However it limits its services to counseling only and does not reason with evidence about the human's change of behavior based on information about their activities.

The agent-based approach to diagnostic decision-making presented in [17] has the potential to support the medical professionals in improving their diagnostic assessments of patients. The purpose is to guide the user in the diagnostic reasoning and support learning and a change of behavior. However, the user's active participation in the dialogue is limited.

To summarize, the most common Agent roles, and consequently, the purposes of the exemplified dialogue systems are the following: i) promoting the human actor to maintain control, learn and change behavior, ii) to guide in reasoning, decision-making and other activities, and iii) to inform in an emphatic and patient way, and behave in a way tailored to the human agent's pace, needs and preferences.

The canonical model of an agent described by Fox and coworkers [3] is an extension of the Belief-Desire-Intention (BDI) model of an autonomous agent (e.g., [2]) and provides formal semantics for the concepts argumentation, decision-making, action, belief, goal (desire) planning (intention) and learning for agents. Literature describes implementation in systems, however, mainly for multi-agent reasoning and decision making without active participation of a human agent in the process [3]. To our knowledge, this model has not served as a base for dialogues between software agents and human agents beyond simple alerts or providing decisions.

The architecture of the agent-based ambient assistive living system presented in [7, 18] integrates knowledge repositories, developed using ACKTUS (Activity-Centered modeling of Knowledge and interaction Tailored to USers), which is a platform for end-user development of knowledge-based systems in the medical and health domain. These repositories are built on semantic models represented using RDF/OWL, which were in [19] extended to integrate components vital for enabling the agent dialogues, based on the canonical model for agent architectures presented by Fox and coworkers [20]. This work led to a multi-agent system, which simulates a dialogue between an agent representing a physician who is the user of the system and who selects a diagnostic hypothesis in a patient case, and an agent representing a domain expert [17]. The domain knowledge and patient knowledge was used for creating arguments, based on sources with different strengths, which the agent-system evaluated for providing the user support in diagnostic decision-making. The domain knowledge was in this case retrieved from clinical practice guidelines, consensus guidelines, and other medical sources.

The ACKTUS core ontology was further extended to incorporate information about a user's environment, for the purpose of forming a user model, which an agent can use for adaptation of support in daily activities [**?**].

An initial design and implementation of an agent-based dialogue system where an agent communicates with the end user in a home environment on health-related topics is presented in [21]. The agent's belief base was created by obtaining information from the ACKTUS knowledge repositories, which had been created by domain experts in the rehabilitation domain [10]. The knowledge consisted of both factual knowledge related to a user, which was used for creating a baseline user model, and procedural knowledge, in the form of rules obtained from domain knowledge repositories. Moreover, interactive templates, or protocols for assessment of different aspects were retrieved from the domain knowledge repositories. The knowledge had been structured and implemented by the domain experts, to be used as assessment protocols by therapists in clinical interviews, or in human-agent dialogues.

The counseling dialogue system discussed in [16] also uses an ontology to represent the user model and the domain knowledge. The domain knowledge is based on behavioral medicine. Their approach consists of six models: theory model, behavior model, protocol model, user model, external data model and a task model. The authors have themselves developed the ontology based on a review of concepts in behavioral medicine, prior work, and discussions with experts in behavioral medicine, and computerized interventions. Their task model consists of the dialogue activity information with dialogue actions. However, the domain experts are not directly involved in the modeling of the agent's knowledge in this system therefore it lacks features for modifications such as future changes in domain knowledge and also it lacks the representation of goals which drives the behavior of the dialogue system.

Only a few applications developed for behavior change integrated into the system semantic models grounded in theories relevant for behavior change [1].

In argumentation literature there are several theoretical frameworks, which aim to formalize dialogues of different types for multi-agent systems. Typically, these adopt the approach of *dialogue games*, since they aim at selecting a "winner" of a dialogue, and then the game rules aim to create the dialogue game a fair competition (e.g., [22]). Based on argumentation schemes, dialogues can be built, e.g., for reasoning about actions to make related to the value of conducting a particular action [23]. Theoretical frameworks have been formalized also for handling nested dialogues of different types (e.g., [22]). However, these are typically not tested with dialogue information from real situations, and consequently, their applicability in situations such as our use case remains to be evaluated.

The base for interaction in [21] was implemented as information-seeking dialogues through which the user can inform the system about preferences, priorities and interests. Based on this information, the system provides feedback to the user in the form of: 1) suggestions of decisions, 2) advices and 3) suggestions of actions to make for obtaining more knowledge about a situation. However, to accomplish other types of dialogues, which are less structured than the *information-seeking* dialogues, an agent architecture is needed, which allows the agent to plan the moves based on e.g., the purpose of the dialogue, the situation, the knowledge available, etc. Therefore, this paper extends earlier work, by developing the components for constructing an intelligent software agent that interacts with a human in a more autonomous, adaptive and purposeful way.

Typically, a formal argumentation framework provides the semantics for solving conflicting arguments for or against a particular topic, regardless what the topic is. The effort to establish an *Argument Interchange Format* (AIF) for exchanging arguments on the World Wide Web is such example [24]. AIF is presented as an ontology, which takes different types of argumentation schemes, scheme-implementation nodes and their relations into consideration, without relating these to a concept-node system for topics. Most frameworks also do not consider vagueness, or uncertainty, or how these factors may change depending on the context of reasoning. Some theoretical work on argumentation frameworks consider *preferences* and *audiences* (e.g., [25–28]), which relate to the context of reasoning. In the presentation of AIF it is mentioned that the nodes may incorporate values, e.g., representing uncertainty, and domain knowledge, but not how such information is included in a formal way. In this work, there is a need to take such factors into consideration in dialogues. Therefore, AIF is applied, and extended with domain knowledge in the form of a concept-node system and templates for assessments of features with associated sets of allowed values. The approach to argumentation dialogues is still generic, since the knowledge domain can be replaced, depending on the topic of interest. This will be illustrated in the pilot study, where the agent retrieves topics from different domains of knowledge, represented as different domain knowledge repositories within the ACKTUS architecture.

Add Hunters articles [**?**,**?**]: e.g., our boolean "do you agree?" compared to Hunters probabilistic degrees of beliefs.

4 Model Based on a Scenario

Our persona called Eva shares similarity with some of the participants in a study conducted by Lindgren and Nilsson [29], and is therefore considered representative. Eva is 84 years old, suffers from pain in her back and legs and had suffered from few falls before a hip fracture. We envision that Eva begins to walk around nighttime, and that she may discuss the situation and her sleep with a nurse. The nurse asks a few specific questions about Eva's activities and health, and this dialogue will wander from one aspect to another, sometimes coming back to a topic already mentioned.

This example of a natural dialogue is rather different from dialogues described in literature, which aims at reaching one particular goal of a dialogue, e.g., [30]. Following the categorization of dialogue types, described by Walton and Krabbe [8], the dialogue with the nurse is a simplified example of a combination of different goals: finding information (*information-seeking* type), generating new knowledge, i.e., conclusions (*inquiry dialogue* type) and deciding upon actions to make (*deliberation* type). In case one of the agents has reasons for arguing for one particular action to be made with the purpose to convince the other, e.g. for safety reasons, the dialogue may include a *persuasive* part, e.g., to convince Eva that she needs to go to the hospital for investigation. Based on this, we can define a set of *generic goals* for the agent to use in its organization of dialogues. More concretely, the generic outcome of each type of dialogue is the following: *information*, *new derived knowledge*, *plan of actions* and a *change of priority.* It may be that all of these types of dialogues need to be conducted to fully explore a particular *topic.* These generic goals will correspond to actions defined in Section 5 and *schemes* described in Section 5.2.

Consequently, the agent needs to be able to handle nested multi-purposed dialogues with different topics. To accomplish this, the agent needs to be able to distinguish between *topic*, *generic goal* and have a semantic model of how these inter-relate in a particular situation. For instance, if the agent would have the dialogue with Eva instead of the nurse, the agent needs a semantic model for how walking around nighttime relates to sleep patterns, pain, cognitive ability, medication, worries, etc. (i.e., a *domain model*). Moreover, it needs strategies to plan next moves, based on a knowledge model, which may not provide a predefined hierarchically organized plan of actions based on goals and sub-goals to be followed, but rather a collection of prioritized actions, among which the order may become determined and changed by the dialogue evolvement and Eva's line of thinking (enabling *situatedness* of dialogues and the agent to adapt).

The *topics* relate to the human agent's goals and priorities. From the software agent's perspective, goals relate to finding answers to questions, finding and deciding about actions to make to increase levels of satisfaction, evaluate actions made, etc. Consequently, for the human agent, goals are purposeful, topic-driven and context dependent.

Therefore, the agent in our approach combines a generic goal with a specific topic, to identify the *specific goal* with an action. In the following we give an example of a dialogue, where the topic selected by the human agent is "Sleep pattern may be disturbed", which is a claim representing a belief, which has not been verified. This is an example of the initiation of an *inquiry* dialogue, aimed at finding and verifying new knowledge. The agent can choose to conduct a dialogue with the purpose to find information about sleep (specific goal) and initiate an information-seeking dialogue. Topics related to sleep are e.g., pain and medication, and consequently, in this situation, the sub-goals are to *find information about pain* and *medication*, respectively. This is an example of how different types of dialogues can be nested, for the purpose to feed information and knowledge into the cooperative reasoning process of solving the overall topic, here chosen by the human actor.

The following is an example of a part of a dialogue, generated by the prototype system in collaboration with Eva, where CA represents the Coach Agent:

1. Eva: Hi Coach, I wonder if not my "Sleep pattern may be disturbed".
2. CA: Hi, Eva. Okay, then I would like to ask about *sleep*. Did the sleeping pills work for you tonight?
3. Eva: No.
4. CA: Why did you not sleep well last night?
5. Eva: Needed to pee and pain.
6. CA: I see. Now I would like to ask about *pain*. Where do you have pain?
7. Eva: In my back.
8. CA: How severe is your pain?
9. Eva: Severe.
10. CA: This does not sound good.

To summarize, the scenario emphasizes the need for the following three semantic models for enabling human-agent dialogues: 1) a *domain model*, which contains generic knowledge about a particular domain, 2) a *user model*, which contains the collected knowledge about the human agent, and 3) a *dialogue activity model*, which provides the relations between topics, generic and specific goals and actions to be made by the participating agents in a dialogue. Moreover, the third model can be seen as the generic behavior knowledge model of the Coach Agent, however, enriched with specific knowledge, which relates only to the Coach Agent, forming the fourth model: 4) an *agent model* (Figure 1). The emphatic components of the example dialogue we regard as generic behavior knowledge, which needs to be present for the dialogue to become pleasant to the Human Actor. This is an example of knowledge, which the agent model needs to encompass.

Finally, the need for *adaptation strategies*, which enable dynamic dialogues following the human agent's line of thinking is illuminated by the scenario.

The user and domain models are extracted from the ACKTUS ontologies [31, **?**] in the dialogues between human and software agents. It should be noted that the domain knowledge repositories are structured in a way so that there are knowledge, which can be considered common, and other which is advanced expert knowledge, corresponding to a medical expert's knowledge. The Coach Agent integrates the

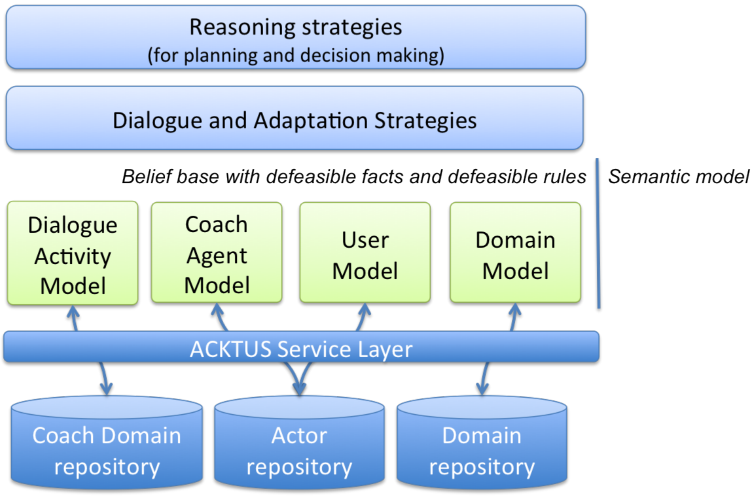

Fig. 1 The agent architecture, containing the Coach Agent's belief base consisting of four models, and its different sources of knowledge.

part that is common knowledge into its belief base in dialogues in our implementation. The dialogue activity model and an initial model of the Coach Agent have been developed as a part of this work, and presented in the following sections.

5 A Model of the Dialogue Activity

As described in Section 4, and following the second requirement listed in the Introduction, the software agent needs to share a common semantic model with the human agent, to be able to reason and decide upon which actions are valuable to the human agent. In this section the dialogue activity model will be defined.

The activity-theoretical model of human activity was used for organizing actions and their goals at different levels [9]. Activity Theory captures the complexity in human activity, including human *needs and motives* as driving force, *goal-directed actions*, and *operations*, which constitute basic actions conditioned by the agents and the environment.

Figure 2 provides an overview of the shared model of a dialogue activity in which both a human and software agent participate. As can be seen, in the dialogues a common *topic* is the representation of the overall *motive for an activity.* Each dialogue is initiated by one of the agents, by posing a selected topic to the other agent.

The identified operations relate to passing information to and receive information from the other agent(s) (*send* and *receive* in Figure 2), and wait for responses.

In addition to these basic and top levels of activity, a set of potential generic sub-actions have been defined, which may take place in the conduction of a dialogue. These are also organized in a hierarchical representation, since an action may serve as a sub-action to more than one action at different levels. However, we identify the following generic set of actions at the highest level: *information-seeking, inquiry, deliberative,* and *persuasive*, related to the common goals of different

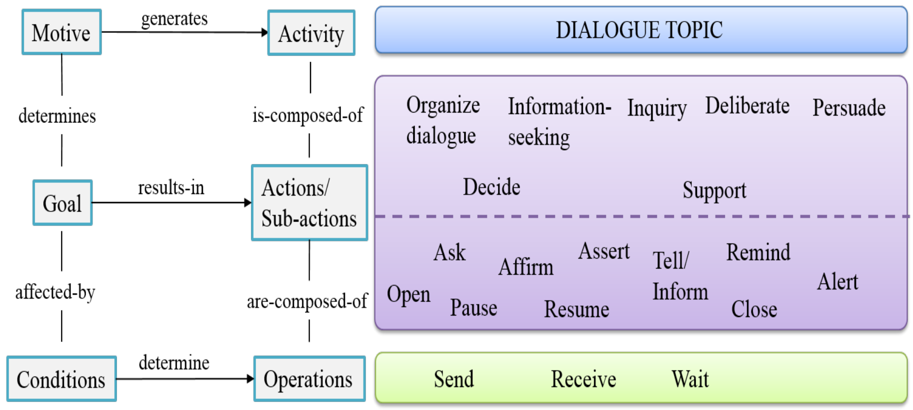

Fig. 2 A model of the dialogue activity, which both a human and a software agent are expected to comply with.

types of dialogues described in Section 4: seek information, find new knowledge, decide upon actions to be made and lead the other agent to change opinion. To these actions, we add the action *organize dialogue*, which contains the sub-actions typical for multi agent dialogues, e.g., *open* and *close*. These sub-actions are called *moves* in multiagent literature [30]. We extend the set of moves (i.e., sub-actions) and include the following moves as valid actions for the agent to take as part of the different actions: *open, close, pause, resume, ask, assert, affirm, inform, remind* and *alert*.

The model of dialogue activity presented in this section defines the actions, which need to be common between the participating agents as a part of a common semantic model. In addition to these, there are tasks, which are conducted by the agent "internally" during a dialogue, which also relates to the execution of a dialogue. These tasks are aimed to handle interruptions in a dialogue, cases when the human agent does not respond, etc. Other tasks are creating and updating the agent's own belief base, decide about what move to make next, which dialogue to initiate, etc. A description of these tasks included in the generic organization of a dialogue mentioned in Figure 2 is provided in Section 6, while the tasks included in the collaborative assessment and decision-making conducted as a part of human-agent dialogues are described in the following sections (Figure 1). To distinguish the lower level actions, which represent different types of moves in a dialogue, we hereafter call these *moves*, explored in the following sub-section, and denote the different dialogue types at a higher level as *actions*, specified in Sub-section 5.2.

5.1 Dialogue Moves

Formally, a dialogue move for the human agent or software agent in this work, is a tuple (t, a, m) where t is timepoint of the dailogue move, a is the agent and m is the dialogue move. The set of dialogue types (d) includes the following: information-seeking (is), inquiry $(wi$ or $ai)$, deliberate (dd), persuasive (pd) and support dialogue (sd). Based on this (t, a, m) tuple, we define the actions as follows (Table 1).

Table 1 Valid actions, i.e., moves and their formats. All moves contain the time identifier t_n and a_i, the identification of the agent, which performs the move.

Move	Form	Comment
Open	$(t_n, a_i, open(d_o, \alpha_m))$	α_m is the topic of a dialogue
Close	$(t_n, a_i, close(d_o, \alpha_m))$	α_m is the topic of a dialogue
Ask	$(t_n, a_i, ask(CQ))$	CQ is a structured question
Affirm	$(t_n, a_i, affirm(\alpha_m))$	α_m is a confirmative expression
Tell	$(t_n, a_i, tell(\alpha_m))$	α_m is the message, typically an advice or information
Remind	$(t_n, a_i, remind((\alpha_m, G), a_j))$	α_m is the reminder, G is the set of reasons for the reminder, and a_j is the agent targeted for the reminder
Alert	$(t_n, a_i, alert((\alpha_m, G), a_j, t_l))$	t_l is the timeout for the action
Assert	$(t_n, a_i, assert(\alpha_m, G))$	α_m is the claim and G is the set of grounds for the claim
Believe	$(t_n, a_i, believe(\alpha_m, CQ))$	α_m is the claim or message and CQ is a structured question

The *Open* move is the first action carried out to initiate a dialogue, and the *Close* move is used for closing, or stating the end of a dialogue. In our implementation, the human initiates the first dialogue, while the agent initiates all sub-dialogues. Moreover, we limit the type of dialogues to *asymmetric* dialogues, where the human is not providing own claims, only responding to questions.

When the agent needs to obtain information from the human actor, it uses the *Ask* move. It fetches the relevant questions to be asked from the ACKTUS knowledge repository and stores them in its knowledge base. The human's answer is a *Tell* move. The agent uses the Tell move for mediating advices or other information, which are not reminders or alerts.

The *Remind* move is used by the agent to remind the human to act. The *Alert* move is also similar to the remind move but with a timeout. It recommends the human to take immediate action in a critical situation, for example, if Eva has forgotten to take medication after breakfast, then the agent sends the alert about taking medicine. The organization of the agent's supportive actions and pro-active behavior is part of the implementation of interventions and is subjected to future work.

The *Assert* move is used for making a claim about some topic, and it is supported by the set of grounds G on which the claim is being based. The claim is a defeasible fact.

The *Affirm* move is used to acknowledge the other agent and its expressions in a more generic way. The typical purpose is to make the other agent comfortable, providing a "fill" in the dialogue.

The *Believe* move is a particular question posed together with an assert move for investigating if the other agent agrees upon the claim, stated in the assert move. The response is given by a tell move and can be either *I agree* or *I disagree* in our implementation (Section 6). However, this measure of the human's belief in the statement could also be based on probabilities, as applied in Hunter's examples [**?**,**?**].

5.1.1 Rules for Dialogue Moves

There are rules related to how the moves can be applied. In dialogue games, the restriction that agents take turn in the dialogues, and are allowed to perform only one move at the time does not apply in our dialogues. The reason is that the agents are not competing for "winning" the dialogue, and natural dialogues do not follow this restriction.

Other restrictions apply, such as the open move for a particular dialogue needs to take place before a close move. Similarly, in case a pause move is done, then no move can be done within this dialogue until the dialogue is resumed by a resume move.

Only one dialogue can be active at a particular time point. This means that when a new dialogue is opened by an open move, and if there is an ongoing dialogue, then this ongoing dialogue is paused. This follows the notion of "focus shifts", which occur when humans conduct activity, a phenomenon used for evaluating interactive systems.

Since it is the Coach Agent, which is directing the ongoing dialogue, the agent is also closing the dialogue. However, the human agent can close the main dialogue, and consequently all sub-dialogues, by switching the topic of the main dialogue.

Some moves have the purpose to defer the responsibility for acting to the other agent. Such moves are *ask*, where the asking agent expects a response, and *alert*, when the alerting agent expects the other agent to take action since the situation requires action. Similarly, whenever the human agent respond to an ask move by a tell move, the responsibility to act is passed to the Coach Agent. Consequently, the agent needs strategies to handle the situation when the other agent does not respond as expected.

In the case when the human agent cease to respond, the Coach Agent puts the dialogue on hold, and defines and stores, if present, partial results of opened dialogues. At a later occasion, the agent can offer the human to resume a dialogue put on hold.

5.2 Goal-Oriented Dialogue Actions

The higher-level actions in Figure 2, which relate to different types of dialogues, can be nested to meet sub-goals in the process of achieving the overall motive for the dialogue activity defined by the topic. In addition, each nested dialogue is initiated by posing a topic, in the same way as the main dialogue is initiated. As a consequence, the execution of the dialogue body needs to handle the different types of dialogues, motives, their outcomes, the organization of these in the dynamic way, which is needed for the agent to be adaptive and flexible. In this process, some constructs are useful for formalizing the dialogues and their outcomes, and for organizing the process of reaching decisions about e.g, actions to make, and their reasons (arguments). The different types of dialogues, their goals, topics and allowed moves are summarized in Table 2.

The purpose of *argumentation dialogues* is to collaboratively compare different views and generate conclusions about which one is best in a situation. This can be what conclusion to draw, what new knowledge do derive, what action to make, and

Table 2 Dialogue types and their characteristics. The topic is drawn from the ACKTUS repositories, using the semantic characteristics of the knowledge nodes.

Type	Goal	Topic	Valid moves
information-seeking	collect information	*concept*	open, ask, tell, affirm, close
inquiry	create new knowledge in the form of defeasible facts or defeasible rules	*i-node* (defeasible fact) or *s-node* (defeasible rule)	open, assert, believe, affirm, close
deliberation	decide about actions to be taken	*i-node* (defeasible fact) or *s-node* (defeasible rule)	open, assert, believe, affirm, close
persuasion	change a priority or belief	*i-node* (defeasible fact) or *s-node* (defeasible rule), in particular their *value* as a part of a *scale*	open, assert, believe, affirm, remind, close
support	enhance human agent's ability	*information-node* or *conclusion*	open, affirm, tell, alert, remind, close

what changes of priority, or beliefs can be achieved. This is a decision-making approach, which provides strategies for handling conflicting information and views. Information, inference rules and conclusions are typically considered *defeasible*, which means that they can be challenged (*attacked*) and defeated. In our work, we assume that all information is defeasible, and no rules are strict rules, valid in all circumstances. Arguments can be attacked in three ways: on their *premises*, on their *inference* (example in Table 4, row 23) and on their *conclusion* (example in Table 4, row 19). In argumentation literature the notion of *argument scheme* is applied for providing semi-formal or formal templates and defeasible inference rules for different kinds of dialogues [32]. A common example is the scheme "Argument from a position to know", which is the starting point for both the human and software agents in our work. Consequently, the agents are considered equally knowledgeable when evaluating their arguments.

Sometimes *critical questions* are defined and associated to a particular scheme, which typically are pointers to counterarguments following Prakken's definition [33]. For the purpose of formalizing the different types of dialogues, the concepts *scheme* and *critical question* are applied and modified. It should be noted, that the evaluation of arguments or the application of schemes alone is not sufficient for evaluating a dialogue. The whole process needs to be completed [33]. In the following, the common features of argumentation are described, which will be followed by a number of subsections in which each type of dialogue is defined. However, we leave the full definitions of the formal argumentation framework applied in our dialogue system for future work.

5.2.1 Argumentation

The topic of dialogues is retrieved from a domain ontology, which incorporates both the Argument Interchange Format (AIF) [24] and the domain knowledge. The domain ontology contains a concept-node system with concepts and their relations. The information-seeking dialogues have a concept as topic, which makes this type of dialogue very generic. Moreover, the domain ontology contains a kind of information node, which has a concept but no values, and functions as the topic for the supportive dialogues (Table 2). We treat these as *statements* in the dialogue, and *literals* in the logic implemented in the dialogue system.

AIF distinguishes between *i-node*, a kind of information node, which in this approach is associated to both a concept and a value, and which form statements; and *s-node* (scheme-node), from which rules are generated. These AIF concepts are used as topic in inquiry, deliberation and persuasion dialogues, with some differences. The formal distinction made in this approach between the inquiry and deliberation dialogues is that the *claim*, i.e., i-node, is in deliberation dialogues associated to a concept related to the node *activity and participation* in the domain ontology, while in the inquiry dialogues the concept is anything else than a node within the activity and participation ontology. Persuasive dialogues can have both kinds, since the focus is the *evaluation of* the phenomenon represented by the concept. This evaluation is represented by its *value* (a measurement of preference or confidence), which is targeted to be changed.

Formally, we treat the i-nodes associated to a value as predicates of the form XXX. Rules extracted from the s-nodes are formally defined as follows: xxx. The rules together with defeasible facts (statements) fulfilling premises of the rules are used for building arguments of the form (x, a), where a is the claim and x is the support for the claim, consisting of rules and facts. An argument is posed in the dialogue with an assert move.

In our dialogue system attacks by the human agent on arguments are identified by a disagreeing response given by the human to the question posed by the agent using the believe move (example in Table 4, row 22-23). This disagreement is respected, and the argument is considered defeated. If the human supports the argument, the argument is considered accepted and validated (example in Table 4, row 24-25). The two arguments in the examples both address actions to address a problem, and as such they are not in conflict with each other. In the health domain we describe in this paper arguments are often not in direct conflict with each other. They may however support decisions or actions, which are beneficial, or important to different degrees. Moreover, as in the example, a defeated argument may be interesting in the particular situation where it was defeated, since it may provide information, which needs to be used for resolving a problem. Consequently, the state-of-the-art approaches to formal argumentation in research literature where one single winning argument is to be identified are insufficient for complex healthcare situations.

A *dialogue line* is the sequence of moves conducted by the agents and their time points (e.g., [30]). Such sequence is visible in the example of a dialogue, presented in Table 3 and Table 4. In the following sections each type of dialogue is further described, and exemplified.

5.2.2 Inquiry Dialogues (wi and ai)

The inquiry dialogue is distinguished from other dialogues in that it is divided into two types following the approach in [30]: *warrant inquiry* (wi) dialogue and *argument inquiry* (ai) dialogue. The topic of a warrant inquiry dialogue is a defeasible fact, while the topic of an argument inquiry dialogue is a defeasible rule. The purpose of the first kind is to create new knowledge, and for the second kind is to create arguments. Consequently, the outcome can be either a conclusion, or an argument. In the case more than one conclusion is generated, a conflict occurs.

The example dialogue shown in Table 3, starts out as a warrant inquiry (wi) dialogue, since the main purpose is to find out whether there is a sleep disorder. The inquiry dialogue evolves when different hypotheses are evaluated and can be assumed to take place after Row 15 in Table 3.

Table 3 Example of different types of dialogues unfolding in a dialogue line.

t_i	Agent	Statement	Formal representation of a move
1	Eva	Hi Coach, I wonder if not my *Sleep pattern may be disturbed.*	$(t_1, a_1, open(wi_1, i\text{-}node_1))$
2	CA	Hi, Eva. Okay,	$(t_2, a_2, affirm(info\text{-}node_1))$
3	CA	Then I would like to talk about *sleep.*	$(t_3, a_2, open(is_1, concept_1))$ $(t_3, a_2, pause(wi_1, i\text{-}node_1))$
3	CA	Did the sleeping pills work for you tonight?	$(t_3, a_2, ask(CQ_1))$
4	Eva	No.	$(t_4, a_1, tell(i\text{-}node_2))$
5	CA	Why did you not sleep well last night?	$(t_5, a_2, ask(CQ_2))$
6	Eva	Needed to pee and pain.	$(t_6, a_1, tell(i\text{-}node_3))$, $(t_6, a_1, tell(i\text{-}node_4))$
7	CA	Okay, I see.	$(t_7, a_2, affirm(info\text{-}node_2))$
8	CA	Now I would like to ask about *pain.*	$(t_8, a_2, open(is_2, concept_2))$ $(t_8, a_2, pause(is_1, concept_1))$
9	CA	Where do you have pain?	$(t_9, a_2, ask(CQ_3))$
10	Eva	In my back.	$(t_{10}, a_1, tell(i\text{-}node_5))$
11	CA	How severe is your pain?	$(t_{11}, a_2, ask(CQ_4))$
12	Eva	Severe.	$(t_{12}, a_1, tell(i\text{-}node_6))$
13	CA	Okay, does not sound good.	$(t_{13}, a_2, affirm(info\text{-}node_3))$
14	CA	Let us go back to the topic *sleep.*	$(t_{14}, a_2, close(is_2, concept_2))$ $(t_{14}, a_2, resume(is_1, concept_1))$
15	CA	How often do you need to get up each night?	$(t_{15}, a_2, ask(CQ_5))$

There is a large number of argumentation schemes defined for reaching new knowledge, representing different reasoning strategies and the confidence in the actor, e.g., argument from expert opinion, argument from a position to know, etc [32]. Some reasoning strategy definitions mirror the range of logical inference strategies, e.g., deductive and inductive reasoning, causal reasoning etc. Consequently, the agents can apply different strategies, depending on the purpose and quality of the available information. For our purposes, we assume at this point that the agents apply an abductive reasoning method, which includes possibilistic values. The following is the scheme defined for abductive argumentation:

Example 1 Abductive argumentation scheme: F is a finding or given set of facts, E is a satisfactory explanation of F, and no alternative explanation E2 given so far is as satisfactory as E. Then E is plausible, as a hypothesis.

This Abductive argumentation scheme example can be applied to our human agent example as follows:

Example 2 The human agent is walking around nighttime (F), the human agent's severe pain (E) is a satisfactory explanation of F, and no alternative explanation given so far is as satisfactory as E, therefore, E is plausible as a hypothesis.

To execute this type of dialogue, the agent first creates a domain model of how different conditions such as pain or incontinence affect the quality of sleep in general. The following is an example, where $\rightarrow$ is used for representing a defeasible causal relationship: ((pain, severe) $\rightarrow$ (disturbed sleep, probable)). The agent combines this knowledge with the knowledge about the human agent represented in the user model, which relates to manifested conditions or observations and the user's priorities: (pain, severe), (walking nighttime), (maintaining pain at endurable levels, highest priority), (maintaining good sleep, highest priority).

However, since there are other potential hypotheses, which may be generated, the Coach Agent needs to continue the reasoning and dialogue to reach a sufficiently complete view of the human agent's situation, following an *exhaustive* method for investigation. The situation is common when there may be more than one hypothesis with same level of satisfaction as the explanation of a finding. Consequently, we will adapt the scheme for allowing the agent work with more than one hypothesis in parallel, and generate an outcome of the inquiry dialogue, which can be utilized as base for further actions. For instance, in the situation when two or more explanatory hypotheses are present with same level of satisfaction, this may lead to the initiation of an information-seeking dialogue, or a deliberation dialogue about what actions to take to resolve the situation with conflicting hypotheses, e.g., involving a healthcare professional (e.g., Table 4). One strategy to resolve the conflict is to accept more than one argument as explanation, and pursue deliberation dialogues for addressing all in a treatment plan.

5.2.3 Deliberation Dialogues (dd)

Argumentative reasoning with the goal to decide about what actions to make is often denoted *practical reasoning* in literature (e.g., [23]). The following scheme for practical reasoning is defined by Walton [34]:

Example 3 In the current circumstances R, we should perform action A, which will result in new circumstances S, which will achieve goal G, which will promote value V.

In our scenario, this may correspond to the following:

Example 4 We know that the pain is severe (R), and if the pain is reduced (A), this will result in the new circumstance where the pain is mild (S), which will achieve the goal to keep the level of pain at a manageable level (G), which promotes good sleep (V).

The execution of this type of dialogue follows the same procedure as in previous example. The topic of the dialogue is *reduce pain*, which is an action (Table 1). The agent creates a domain model of how pain and other conditions and medication may affect the quality of sleep in general: ((pain, severe) $\rightarrow$ (disturbed sleep, probable)), (painkiller $\rightarrow$ reduced pain). The agent combines this knowledge with the knowledge about the human agent represented in the user model:

Table 4 Brief example of persuasive and deliberation dialogues.

t_i	Agent	Statement	Formal representation of a move
14	CA	Let us go back to the topic *sleep* and *priority*.	$(t_{14}, a_2, close(is_2, concept_1))$ $(t_{14}, a_2, open(pd_1, i\text{-}node_7))$
15	CA	You said that maintaining good sleep is of highest priority.	$(t_{15}, a_2, remind(i\text{-}node_8, \{\}), a_1)$
16	Eva	Yes.	$(t_{16}, a_1, affirm(info\text{-}node_4))$
17	CA	You also told earlier that maintaining pain at endurable levels is not important.	$(t_{17}, a_2, remind(i\text{-}node_7, \{\}), a_1)$
18	Eva	Yes.	$(t_{18}, a_1, affirm(info\text{-}node_4))$
19	CA	Since it is likely that severe pain causes disturbed sleep, and you have severe pain, and you think maintaining good sleep is of highest priority, then it should be important to maintain pain condition at an acceptable level. What do you think?	$(t_{19}, a_2, assert(i\text{-}node_9, \{s\text{-}node_1, i\text{-}node_6, i\text{-}node_8\}))$ $(t_{19}, a_2, ask(CQ_6))$
20	Eva	Okay, important I guess.	$(t_{20}, a_1, affirm(info\text{-}node_1))$ $(t_{20}, a_1, assert(i\text{-}node_9, \{s\text{-}node_1, i\text{-}node_6, i\text{-}node_8\}))$
21	CA	Okay, let us talk about what to do about the *pain*.	$(t_{21}, a_2, affirm(info\text{-}node_1))$ $(t_{21}, a_2, close(pd_1, i\text{-}node_7))$ $(t_{21}, a_2, open(dd_1, i\text{-}node_{10}))$
22	CA	Since taking painkiller typically reduces pain, then you can take painkiller. What do you think?	$(t_{22}, a_2, assert(i\text{-}node_{10}, \{s\text{-}node_2\}))$ $(t_{22}, a_2, ask(CQ_6))$
23	Eva	No, painkiller does not work.	$(t_{23}, a_1, tell(i\text{-}node_5))$ $(t_{23}, a_1, assert(i\text{-}node_{11}, \{s\text{-}node_3\}))$
24	CA	Okay, I see. Do you want to talk to your nurse about medication?	$(t_{24}, a_2, affirm(info\text{-}node_2))$ $(t_{24}, a_2, ask(CQ_7))$
25	Eva	Yes.	$(t_{25}, a_1, tell(i\text{-}node_{11}))$
26	CA	Okay then I wonder if I could *inform nurse about summary* for you.	$(t_{26}, a_2, affirm(info\text{-}node_1))$ $(t_{26}, a_2, close(dd_1, i\text{-}node_{10}))$ $(t_{26}, a_2, open(sd_1, concl_1))$ $(t_{26}, a_2, tell(info\text{-}node_5))$
27	Eva	Yes.	$(t_{27}, a_1, tell(i\text{-}node_{11}))$
28	CA	Okay, don't forget to talk to the nurse about your medication!	$(t_{28}, a_2, affirm(info\text{-}node_1))$ $(t_{28}, a_2, remind(i\text{-}node_{11}, \{s\text{-}node_3, s\text{-}node_1, i\text{-}node_6, i\text{-}node_8\}), a_1)$
29	Eva	Yes. Goodbye!	$(t_{29}, a_1, tell(i\text{-}node_{11}))$ $(t_{29}, a_1, close(sd_1, concl_1))$

(pain, severe), (walking nighttime), (maintaining pain at endurable levels, highest priority), (maintaining good sleep, highest priority).

The dialogue in this situation will deal with what to do to reduce the pain level, in order to affect the sleep disturbance in a positive way. In our example, using painkiller to reduce pain would be the suggestion (rows 21-25 in Table 4).

The outcome of a deliberation dialogue is a plan of action, which may be empty.

5.2.4 Persuasive Dialogues (pd)

Persuasive dialogues aims at resolving a conflict of opinion [33].

To formalize this type of dialogue, the agent creates a user model of the human's prioritized activities, for example: (taking medication, important), (wellbeing, very important) and (maintaining good sleep, highest priority). For illustrating a persuasive dialogue, the user's preference regarding managing pain is set to the the following: (maintaining pain at endurable levels, not important). Then the agent combines this knowledge with the knowledge obtained from the human actor during the dialogue. Suppose the human has disturbed sleep and initiates a dialogue with the Coach Agent about the topic "sleep patterns may be disturbed". During the information seeking dialogue, the agent asks the human about pain and if the human responds that a pain condition is present, and the pain is "severe", then the agent updates its user model with (pain condition, yes) and (pain, severe), and initiates a persuasive dialogue (rows 14-20 in Table 4) based on its knowledge about the relationship between sleep and pain condition.

The agent makes new statements, reminds the human about the relationship between pain and sleep, and the human changes the evaluation of the importance of managing the pain condition (the value *important* is stronger than the value *not important*). Next step for the agent is to propose actions to do something about the situation.

5.2.5 Information-Seeking Dialogues (is)

The topic of an information-seeking dialogue is a *concept*, which is a broader topic than the defeasible facts or rules, used as topics for an inquiry, deliberation or persuasion dialogue. An information-seeking dialogue typically unfolds as an interview, where the Coach Agent in our examples asks relevant questions to the human agent, and receives answers. To some extent the agent evaluates the new information, however, primarily for deciding upon next step, typically what question to ask next. Therefore, we use the move *tell* in the information seeking dialogues instead of *assert*, to distinguish between answering questions for reasoning purposes such as in inquiry dialogues, and answering primarily for collecting information.

5.2.6 Support Dialogues (sd)

The topic of a supportive dialogue is an *information-node*, which is not associated to a value. The content of a support dialogue is typically the outcome of an earlier conducted deliberation dialogue, where the human agent and the Coach Agent have agreed upon a plan of actions to be conducted. The actions performed as support dialogues are one of the following: provide the human agent with information or advice, remind the person of actions to make, and alert the person when important things need to be done. A remind and alert move are arguments, which contain the information about what is to be done, i.e., a claim, together with the motivations, i.e., the grounds, which support the claim.

6 The Coach Agent Model and Implementation

For the Coach Agent to be able to conduct dialogues similar to natural dialogues, the agent was designed as an actor with its own knowledge base, goals and priorities, represented as an ontology, extended from the ACKTUS core ontology. There is for instance, behavioral knowledge, which is generic regardless which type of dialogue or topic is at focus. Emphatic behavior is such generic social knowledge fetched from the agent's repository, since this knowledge needs its own semantic model, yet related to the human actor's, the Coach Agent was modeled as a separate project in ACKTUS (Figure 1). As a consequence, the Coach Agent shares the core ontology with the human agent and the domain knowledge models. This functions as the common vocabulary in dialogues.

The Coach agent performs two types of tasks: *dialogue-specific* tasks and, *internal processing* tasks. These tasks are further explained as follows:

- Dialogue-specific tasks are performed in cooperation with the human agent to enhance the adaptation of the agent's behavior. The Agent model contains structures, which function as tools for the Coach Agent to use in the adaptation of its behavior. These tools aids the agent in the *conceptualization, knowledge acquisition and reasoning*, related to cooperative human-agent activities which in our case is dialogues. (6.3). EXAMPLES
- Internal processing tasks relate to the execution of a dialogue, i.e., the protocols for different dialogues. These tasks are aimed to handle interruptions in a dialogue, situations when the human agent does not respond, etc. It also includes the tasks of creating and updating the agent's own belief base, decide about what move to make next, which dialogue to initiate, etc. (Section 6.4)

6.1 System implementation

The Human-Coach Agent Dialogue(H-CAD) system has been implemented in Java as backend with HTML, JSP, jQuery and AJAX as frontend. The communication between a human and the Coach Agent happens as JSON messages. The H-CAD system is embedded into iHelp application. The iHelp is a prototype application for the older adults.

6.2 Adaptation Strategies

The protocol for conducting dialogues described in Section 5.1.1 contains a limited set of dialogue rules in order to leave room for adaptation strategies. At this point, we adopt strategies based on therapeutic values, however, we envision that these strategies can be further improved and tailored through supplementing with learning methods in the future.

The agent uses the same strategy for selecting the next move in the different types of dialogues. The agent retrieves the set of information, which relates to the topic, and organizes these into a plan following the human actor's preferences among the topics. At each step in a dialogue when receiving new information from the human, the agent analyses the information, decides whether to continue the current dialogue, pause it for opening a new sub-dialogue, or close the dialogue to possibly start a new dialogue. After this, the agent either abandons the current plan to make a new plan, or adjust the current plan.

Depending on the type of dialogue, the content of the plan is different. In the case of information-seeking dialogues, the plan contains a set of questions (outcome is answers); in inquiry dialogues the plan contains either a set of defeasible rules or facts (outcome is arguments or new knowledge); in the case of a deliberation dialogue the plan contains a set of potential actions to be decided upon (outcome is actions agreed upon); and in the persuasion dialogues the plan contains a set of statements and arguments related to the dialogue goal, which may be used in a persuasive argumentation dialogue (outcome is a change of priority or belief).

A balance is strived for between the qualities exhaustiveness and efficiency. In any assessments done in healthcare, the optimal would be a complete set of information in order to optimize decisions. However, some information may not be available, time may be limited and the cooperation of the human actor may not persist if the assessment takes too long time, possibly due to limited motivation or capacity. Therefore, strategies have been built into the Coach Agent to do the following for selecting the next move and reducing the number of moves:

- i prioritize topics following the human's prioritization
- ii prioritize topics following the domain experts' prioritization as structured in the domain knowledge repository
- ii prioritize topics for which the agent has advice or suggestion of intervention, already in information-seeking dialogues
- iii identify and rank questions associated to premises based on the frequency they occur in rules
- iv identify partially satisfied rules, and pose questions to satisfy these before focussing on areas where there are less knowledge
- v identify questions which are of "screening" type, aimed to find indications of problems. This by identifying rules with conclusions associated to concepts at higher levels of specificity, e.g., "memory" rather than "episodic memory"
- vi rely on older information and reduce number of repeated questions (will be based on learning methods in the future, in order to discover how frequent conditions changes in an individual as well as groups of individuals, e.g. pain conditions)

The following is the algorithm for the strategies implemented in the dialogue system.

AAA

6.3 Emphatic behavior

The Coach Agent uses its ontology to associate each action with a concept that is common for both the human and the agents such as *starting, sustaining* and *ending a conversation.* These concepts are used to define certain behaviors that are typically appropriate in a phase of a dialogue, for example, *"Hi"* is associated to concept *"starting a conversation"*. The different types of dialogues and dialogue moves described in previous section are also integrated in the agent's model of dialogue activity embedded in the ontology, for instance, the move *affirm*, means that the agent emphasizes and affirms the emotional state, which the human agent expresses, by using different empathic statements such as *"does not sound good"* and *"I see"*. This enables the Coach Agent to simulate the behavior of a participant who listens with some empathy in our pilot study. The agent distinguishes between positive and negative emotional states in the human by analyzing if the human's response to a question is associated one of the to the categories "normal/healthy" or "pathologic/not normal" in the ontology. For instance, the answer "yes" may be positive or negative, depending on the context. However, not all values are suitable to be associated in this way, and consequently, the agent's ability to respond in an emphatic way in current implementation is limited.

The Coach Agent acquires new knowledge partly by posing explicit questions in the asymmetric dialogues, where the human is merely responding to questions. One example is in the situation when the Coach Agent presents a claim about a topic, followed by the move *believe*, which contains the question *"What do you think?"* (e.g., Line 19 and 22 in the dialogue example in Table 4). This is a generic question, which gives the agent some indication if the human agent agrees to the claim or if the agent needs to adjust its strategies for the next move in the dialogue.

The agent also retrieves formal models from the agent domain repository, which can be used for reasoning, in the same way the agent retrieves models from another domain repository in the dialogues about a topic. EXAMPLE

6.4 Reasoning Strategies?

The algorithm depicted in Figure 3, begins with the assumption that there exists at least one rule in the knowledge base for the agent to fetch during the dialogue. This rule has a predicate which has the dialogue topic as the conclusion (refer equation1).

$$\exists_r \in RQ[\,(\sum_{j=1}^{m} CQ_j, l) \;\rightarrow P] \tag{1}$$

where, r is a rule, which belongs to R,
R is a set of rules in the agents Belief base,
$Q[\,(\,a, b) \;\rightarrow b]$ is a predicate.

6.5 Flowchart II: "In-between dialogues"

6.5.1 Local Look Ahead

Given the current formula F and the current cause bin CB, then the decision heuristic of the coach agent selects a single question CQ and the next cause bin for its next dialogue move. It tests the immediate implications of asking and not asking a question to the human. For each question, there are 'a' number of answers. Each answer has an array with 'c' cause weight values for each cause. When an answer is selected, the coach agent takes the corresponding conclusion weight array and adds it to the cause bin.

Local look ahead method, selects the next question to ask, such that it leads to confirmation of most likely cause, in the cause bin. After each dialogue move, it evaluates the cause bin to select the most likely cause. It then searches for the

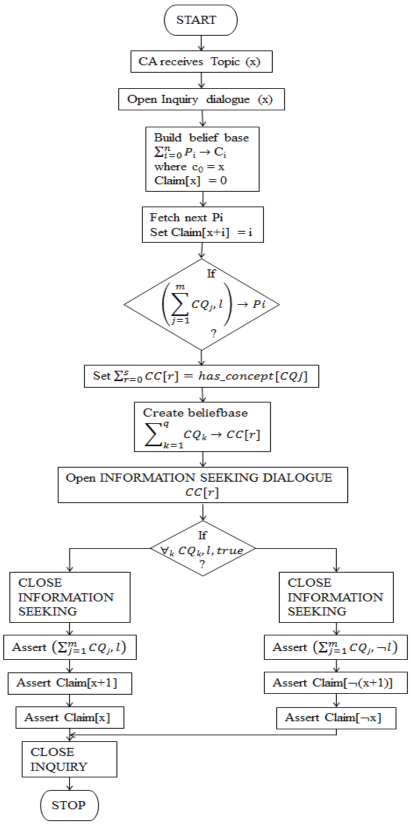

Fig. 3 Flowchart I: "Dialogues": This figure depicts an adaptive dialogue tree, that allows the agent to lead the human through an interactive decision process (DIALOGUE?).

Algorithm 1 In-between dialogues algorithm

1: **procedure** PRIORITIZE–CONFIRMATION/ELIMINATION OF CAUSE
2: Set $\sum_{r=1} {}^{s}CC[r] = has\_concept[CQ_j]$ (from Figure 3)
3: where $\sum_{r=1} {}^{s}CC[r] \ni [CB_1, CB_2, \ldots, CB_n]$ i.e., each Cause Bin (CB) belongs to a concept (CC)
4: create beliefbase

$$F = \sum_{k=1} {}^{q}CQ_k \to CC[r] \tag{2}$$

5: Select next CQ_k such that it confirms/eliminates a CB_n based on lookahead (described in the section 6.5.1)
6: **end procedure**

question which has an answer which leads to confirmation of the cause. If the human selects some other answer than the answer which leads to confirmation of the cause, due to the nature of cause weights assigned to each answer, the evaluated cause bin value will change such a way that current most likely cause will become less likely and some other cause will become most likely. The local look ahead method, repeats the above process to select the next question. This process continues, until a cause is confirmed or all the questions, from the belief base for the selected topic, have been asked. Cause confirmation happens when the most likely cause bin value is above the threshold value w.r.t other cause bin values.

After confirming a cause, the coach agent proposes the same to the human. The human agent can then either agree or disagree to a cause. If the human agrees then the coach agent initiates a deliberation dialogue to come to a solution or action to take for handling the cause. Whereas in the latter case, when the human disagrees to the cause concluded by the agent, the Coach agent continues to maximize the next optimal cause of the dialogue topic. At any point in the dialogue, the human can pause or quit the dialogue.

6.6 Evaluation Study Results

Each participant was first engaged in a baseline assessment, using the iHelp application in dialogue with a therapist. By answering questions in a structured format, information is collected and a baseline user model including prioritized goals can be formed by the agent based on the information.

Next the participant was instructed to freely explore the dialogue system, where (s)he could select topics. When a topic was selected, the Coach Agent was activated and directed the dialogue.

7 Conclusions

The purpose of this work was to define a generic model of purposeful human-agent dialogues about health-related topics. This was done based on analyses of scenarios, personas and models of human behavior. The resulting conceptual model includes four models to be included in a software agent's belief base; i) a user model, ii) a model of the domain knowledge related to the topic of the dialogue, iii) an agent model, and iv) a dialogue activity model. The major contribution of this work is the dialogue activity model to be shared between the human and software agents. The models were implemented into a prototype system for human-agent dialogues, which was evaluated by therapists.

In future work we will develop the agent model further, adding autonomous reasoning and behavior strategies related to supportive actions, which may or may not be mediated in dialogue format. Future work includes also the development of the dialogue and reasoning strategies for the agent to improve its ability to adapt to the individual and the situatedness of contextual and natural dialogues, handle knowledge obtained from different heterogeneous sources, e.g., in a home environment, and reason and make decisions in the presence of uncertain and incomplete information.

8 Acknowledgments

This research was partly funded by the Swedish Brainpower program.

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